

Heterogeneity in Dynare 7 Extensions

Dynare Masterclass 2026

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Four independent extensions. Each builds on the earlier sessions, plus a little glue:

- ① **Reiter (2009)**: a state-space solution from the same residuals.
- ② **Representative-agent models**: the aggregate-only sequence-space Jacobian.
- ③ **Second moments**: read straight off $\text{dr} . G$.
- ④ **Nonlinear transitions**: the MIT shock, still in sequence space.

Roadmap

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→ Some steps read undocumented `oo_.heterogeneity` fields.

Field names and layout are not part of the stable interface; they may move.

Table of Contents

- 1 Reiter (2009): a state-space solution
- 2 Representative-agent models via the aggregate-only solve
- 3 Second moments from the sequence-space Jacobian
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The state vector

Stack the discretised heterogeneous policy, the discretised distribution, and the aggregate vector:

$$S_t = \begin{bmatrix} \mathbf{x}_t^\top, & \Omega_t^\top, & X_t^\top \end{bmatrix}^\top.$$

Three equation blocks, all from the earlier sessions:

$$F(\mathbf{x}_t, \mathbb{E}_t \mathbf{x}_{t+1}, X_{t-1}, X_t, \mathbb{E}_t X_{t+1}, \mathcal{E}_t) = 0,$$

$$\Omega_t - \Lambda(\mathbf{x}_{t-1}) \Omega_{t-1} = 0,$$

$$G(X_{t-1}, X_t, \mathbb{E}_t X_{t+1}, \mathcal{E}_t) = 0.$$

→ The distribution sweep is in predetermined form: Ω_t depends on the lagged policy $\mathbf{x}_{t-1}$.

Why Reiter?

Reiter [2009]: the linearised model in standard linear rational-expectations form,

$$A \mathbb{E}_t dS_{t+1} + B dS_t + C dS_{t-1} + E \mathcal{E}_t = 0,$$

solved by QZ [Sims, 2002]. Recursive law of motion:

$$dS_t = P dS_{t-1} + Q \mathcal{E}_t.$$

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$$dS_t = P dS_{t-1} + Q \mathcal{E}_t.$$

→ First order: certainty equivalence ties column 1 of $\text{dr}.G$ (the perfect-foresight MIT-shock impulse response) to this law of motion.

Built from what is already there

Not a separate toolkit. Reuses almost everything already in place:

- the same .mod file;
- the steady state from `heterogeneity_load_steady_state`;
- the same heterogeneous and aggregate residuals;
- the distribution lottery Λ the sweep step already builds.

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New: two pieces of glue.

- a stacked residual $F(S_{t-1}, S_t, S_{t+1}, \mathcal{E}_t)$, differentiated to (A, B, C, E) analytically or by finite difference;
- Sims [2002]'s QZ for $dS_t = P dS_{t-1} + Q \mathcal{E}_t$.

Two facts the linearisation must respect

- **The distribution is a predetermined state**, exactly as in Dynare's representative-agent solver:

$$\Omega_t = \Lambda(\mathbf{x}_{t-1}) \Omega_{t-1}.$$

The forward-looking objects in S_t are the heterogeneous policies carrying an expectation; they jump with the aggregate shock. QZ pins them down (Blanchard–Kahn).

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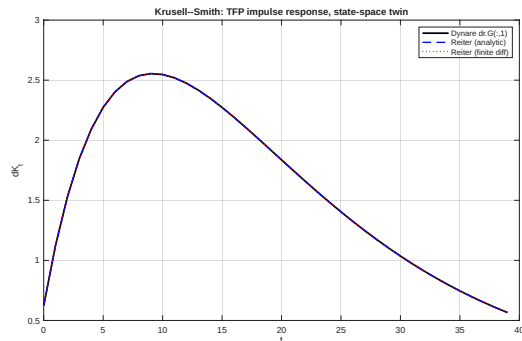
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- **The lottery is policy-dependent.** The expectation brackets the asset choice on the grid, so the bracket weights move with the choice. The time-iteration step already rebuilds them; the Reiter linearisation reuses that rebuild. Feedback captured exactly.

Match against `dr.G`

Capital response to a TFP innovation, Krusell–Smith.



State space matches `dr.G` to $\approx 10^{-9}$ over 40 periods (analytic 6×10^{-10} , finite difference 10^{-9}). Certainty equivalence at first order [Auclert et al., 2021].

Shrinking the state

QZ runs on $S_t = (\mathbf{x}_t, \Omega_t, X_t)$, at a cost cubic in its length.

- Two blocks scale with the grid: the policy array $\mathbf{x}_t$ and the histogram Ω_t (one entry per node, hundreds to thousands).
- Estimation wants one solve per parameter draw, so the state size is the bottleneck.

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→ Goal: describe each block by a handful of numbers instead of every node.

Reduce the distribution: the idea

The histogram is smooth, so represent it by a few cubic-spline coefficients β_t on the asset grid (say 40–80 numbers, not the full node count):

$$\underbrace{\Omega_t \approx P \beta_t}_{\text{rebuild the histogram}}, \quad \underbrace{\beta_t = W \Omega_t}_{\text{read the coefficients}}, \quad WP = I.$$

- P : tall, its columns are the basis shapes on the grid.
- W : short, recovers the coefficients; $WP = I$ means the round trip is lossless on any shape the basis can represent.

Reduce the distribution: the system

Take the linear system $A \mathbf{E}_t dS_{t+1} + B dS_t + C dS_{t-1} + E \mathcal{E}_t = 0$: one column per state entry, one row per equation. Two steps.

- ① **Restrict the unknowns:** $dS_t = \mathcal{P} d\hat{S}_t$, $\mathcal{P} = \text{diag}(I, P, I)$. The distribution columns collapse to β_t . Fewer unknowns, but still one row per node: overdetermined.
- ② **Re-square:** left-multiply by $\mathcal{W} = \text{diag}(I, W, I)$, replacing the distribution rows by one equation per coefficient.

$$\mathcal{W} A \mathcal{P} \mathbf{E}_t d\hat{S}_{t+1} + \mathcal{W} B \mathcal{P} d\hat{S}_t + \mathcal{W} C \mathcal{P} d\hat{S}_{t-1} + \mathcal{W} E \mathcal{E}_t = 0.$$

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Square again, so QZ runs on these smaller matrices; $WP = I$ makes the substitution exact on the subspace. $\rightarrow$ Put the SUM vectors in the basis and the aggregates $\int \bar{y}_t d\Omega_t$ stay exact: $\approx 10^{-7}$ at 80 knots, 1.6×10^{-5} at 40.

The policy block is the catch

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- But the policy array also scales with the grid, and usually dominates the state size, so reducing the distribution alone buys only a modest speedup.
- The policy block is **forward-looking**: Blanchard–Kahn picks its jump to land on the stable path. Force that jump into a fixed small basis and it misses the true direction, so the saddle-path solution goes off (5–30% error, non-monotone in the modes kept).

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→ The predetermined half is done and exact; the forward half is open. Bayer and Luetticke [2020] reduce a value function and succeed; reconciling their endogenous-grid residual with the first-order-condition residual here is the missing step.

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Drop the agents, keep the machinery

heterogeneity_solve feeds on two things:

- the aggregate equations $G(X_{t-1}, X_t, \mathbb{E}_t X_{t+1}, \mathcal{E}_t) = 0$;
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Remove the agents: only the aggregate equations survive. Every standard linear Dynare model qualifies, so the same pipeline returns the sequence-space Jacobian of a textbook representative-agent model.

- Two examples: the small New Keynesian model of Herbst and Schorfheide [2015], the medium-scale Smets and Wouters [2007].

One sparse block-Toeplitz solve

Stack the linearised aggregate equations over $t = 1, \dots, T$:

$$A \mathbb{E}_t X_{t+1} + B X_t + C X_{t-1} + E \mathcal{E}_t = 0, \quad X_0 = 0, \quad X_{T+1} = 0.$$

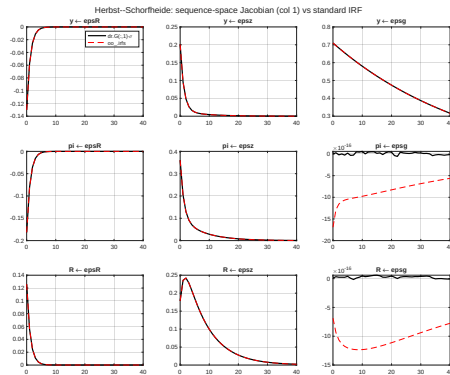
- A, B, C, E : the four slices of the preprocessor's aggregate Jacobian at the steady state. Constant for a `model(linear)` block.
- Stacked over time, this is one banded sparse system; solving it gives the full Jacobian $X = G \mathcal{E}$.
- The same G that `heterogeneity_solve` produces, with the heterogeneous feedback switched off. Helper: `ra_sequence_jacobian.m`.

Certainty equivalence, again

- Column 1 of G is the unanticipated impulse response: it reproduces `oo_.irfs` after rescaling by the shock's standard deviation.
- The other columns are the anticipated (news-shock) responses that fill the $T \times T$ operator: the sequence-space Jacobian of Auclert et al. [2021].
- Truncation T must clear the model's persistence. The forward-looking capital block of Smets and Wouters [2007] needs T well above the plotted window, else the terminal condition $X_{T+1} = 0$ leaks back.

Match against standard Dynare

Sequence-space Jacobian (column 1) against `stoch_simul`.



Herbst-Schorfheide: 7×10^{-15} at $T = 300$. Smets-Wouters: 7×10^{-8} at $T = 600$, machine precision at $T = 1000$.

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`heterogeneity_solve` already produced the impulse responses: the first column `dr.G.X.u(:,1)` is the response of X to a unit innovation in u .

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- An impulse response *is* the moving-average kernel. Write $g_h = \text{dr.G.X.u}(h+1,1)$, so $X_t = \sum_{h \geq 0} g_h \mathcal{E}_{t-h}$.
- With innovations of variance σ^2 , every second moment is a sum of products of those same responses:

$$\text{Var}(X) = \sigma^2 \sum_{h \geq 0} g_h^2, \quad \text{Cov}(X_t, X_{t-j}) = \sigma^2 \sum_{h \geq 0} g_{h+j} g_h.$$

- Several shocks: sum the same products over shocks.

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→ No extra solve, no Lyapunov equation, no simulation: variances, autocorrelations and cross-correlations fall straight out of `dr.G`.

→ Helper `ha_moments.m`; cross-check against a long simulation.

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A nonlinear transition

MIT shock: one unanticipated innovation $\mathcal{E}_0$, then $\mathcal{E}_t = 0$, perfect foresight, return to X^* . Want the exact path $\{\bar{x}_t, \Omega_t, X_t\}_{t \geq 1}$.

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- First order: the path scales with $\mathcal{E}_0$ (certainty equivalence), and is column 1 of $dr \cdot G$.
- Large $\mathcal{E}_0$: the backward operator $\mathfrak{F}$ is nonlinear (kinks, occasionally-binding constraints), so the path is not proportional.

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The exact path is the recursive equilibrium, indexed by time:

$$\bar{x}_t = \mathfrak{F}(\bar{x}_{t+1}, Y_t),$$

$$\Omega_{t+1} = \bar{\Lambda}(\bar{x}_t) \Omega_t,$$

$$G\left(\int \bar{y}_t d\Omega_t, X_{t-1}, X_t, X_{t+1}, \mathcal{E}_t\right) = 0.$$

→ Same $\mathfrak{F}$, $\bar{\Lambda}$, G as in the earlier sessions.

Newton on the aggregate path

Unknown: the aggregate path $X_1, \dots, X_T$, terminal $X_{T+1} = X^*$. Given a guess:

- ① **backward pass** $t = T \rightarrow 1$: $\bar{x}_t = \mathfrak{x}(\bar{x}_{t+1}, Y_t)$, from $\bar{x}_{T+1} = \bar{x}^*$;
- ② **forward pass** $t = 1 \rightarrow T$: $\Omega_{t+1} = \bar{\Lambda}(\bar{x}_t) \Omega_t$, from $\Omega_1 = \Omega^*$;
- ③ **aggregate** $\int \bar{y}_t d\Omega_t$ into $G \rightarrow$ residual $H(X_1, \dots, X_T)$.

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Newton on $H = 0$ [Auclert et al., 2021].

- Its Jacobian is the operator the linear solver inverts for $\text{dr}.G$: the first step from the steady state is exactly column 1 of $\text{dr}.G$, the linear impulse response.
- The remaining steps are the nonlinearity. Helper `ha_td_solve.m`.

The backward step is one time-iteration step

The forward push-forward $\bar{\Lambda}$ and the aggregate block G are reused from the earlier sessions. Only the backward operator $\mathfrak{F}$ is not a Dynare callable.

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- In sequence space [Auclert et al., 2021] the user supplies $\mathfrak{F}$ (the endogenous-grid step).
- Here one application of $\mathfrak{F}$ is exactly one time-iteration step: solve the heterogeneous block

$$\mathcal{F}[\bar{x}_t, \bar{x}_{t+1}, Y_t] = 0 \quad \Longleftrightarrow \quad \text{dynamic_het1_resid} = 0$$

for $\bar{x}_t$ at fixed continuation $\bar{x}_{t+1}$, per node, complementarity via `lmmcp`.

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→ Rebuild $\mathfrak{F}$ from the shipped residual: no endogenous-grid coding, no model-specific inversion. Works for *any* heterogeneous block the preprocessor produces.

→ Helper `ha_backward_step.m`.

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